TopMark Tutors

SACE STAGE 2 · REVISION CHECKLIST

Chemistry

Every topic you need to know, in one checklist.

4 topics | 55 checkpoints | TopMark Tutors

How to use this checklist

Work through every line and tick honestly. Do the whole thing in one sitting before you revise anything — that first pass becomes your study plan for the term.

1. **Revised** — You have gone back over the content and understand it.
2. **Practised** — You have done real exam questions on it, not just read notes.
3. **Confident** — You could do it in the exam, under time, with no notes.

Most students stop at column 1. Re-reading notes feels like studying, but marks are won in columns 2 and 3. Anything ticked in column 1 and blank in column 3 is exactly where your marks are leaking.

Externally assessed by written examination (30% of final grade).

Topic 1 — Monitoring the Environment

Concentration and measurement

CONTENT	REVISED	PRACTISED	CONFIDENT
Calculating concentration in mol L^{-1}, g L^{-1}, ppm and ppb, and converting between them	☐	☐	☐
Dilution calculations ($c_1V_1 = c_2V_2$) and preparing standard solutions	☐	☐	☐
Distinguishing accuracy, precision, reliability and validity in measurement	☐	☐	☐
Identifying random vs systematic error and their effect on results	☐	☐	☐

Analytical techniques

Atomic absorption spectroscopy (AAS): principle, why elements absorb specific wavelengths	☐	☐	☐
Using a calibration curve to determine an unknown concentration	☐	☐	☐
The standard addition method and when it is used to correct for matrix interference	☐	☐	☐
Chromatography (paper, TLC, GC, HPLC): stationary vs mobile phase, polarity and retention	☐	☐	☐
Explaining separation and elution order in terms of relative attraction to each phase	☐	☐	☐

Volumetric analysis

Acid–base titration calculations, including choosing an appropriate indicator	☐	☐	☐
Redox and precipitation titrations, writing balanced ionic equations	☐	☐	☐
Calculating average titre correctly (discarding outliers) and expressing significant figures	☐	☐	☐

Topic 1 — Monitoring the Environment (continued)

Water and environmental chemistry

Dissolved oxygen, biochemical oxygen demand (BOD) and their significance	☐	☐	☐
Eutrophication: causes, the role of nitrate and phosphate, and consequences	☐	☐	☐
Heavy metal contamination and bioaccumulation	☐	☐	☐
Acid rain: formation from SO_2 and NO_x, and its environmental effects	☐	☐	☐

WEAK SPOTS TO COME BACK TO

Topic 2 — Managing Chemical Processes

Equilibrium

CONTENT	REVISED	PRACTISED	CONFIDENT
Features of dynamic equilibrium and writing the Kc expression from an equation	☐	☐	☐
Kc calculations using an ICE table, including determining if a system is at equilibrium	☐	☐	☐
Interpreting the magnitude of Kc (extent of reaction)	☐	☐	☐
Le Châtelier's principle: effect of concentration, pressure, volume and temperature	☐	☐	☐
Using a Kc-vs-temperature graph to deduce whether a reaction is exothermic or endothermic	☐	☐	☐
Why a catalyst changes rate but not Kc or equilibrium position	☐	☐	☐

Rates of reaction

Collision theory and activation energy	☐	☐	☐
How temperature, concentration, surface area and catalysts affect rate	☐	☐	☐
Catalysts: providing an alternative pathway with lower activation energy	☐	☐	☐

Industrial processes

The Haber process: conditions used and why they are a compromise between rate and yield	☐	☐	☐
The Contact Process for sulfuric acid, including the role of V_2O_5	☐	☐	☐
Explaining industrial conditions in terms of economics, safety and environmental impact	☐	☐	☐
Percentage yield and atom economy calculations	☐	☐	☐

WEAK SPOTS TO COME BACK TO

Topic 3 — Organic and Biological Chemistry

Structure and nomenclature

CONTENT	REVISED	PRACTISED	CONFIDENT
Naming and drawing alkanes, alkenes, alkynes, haloalkanes and aromatic compounds	☐	☐	☐
Identifying functional groups: alcohol, aldehyde, ketone, carboxylic acid, ester, amine, amide	☐	☐	☐
Structural isomers and drawing full structural formulae	☐	☐	☐

Reactions of organic compounds

Addition, substitution, oxidation, esterification and hydrolysis reactions	☐	☐	☐
Predicting products and writing equations for multi-step synthesis	☐	☐	☐
Reagents and conditions required at each step of a synthesis	☐	☐	☐
Distinguishing tests (e.g. bromine water, Tollens' reagent, iron(III) chloride)	☐	☐	☐

Biological molecules

Amino acids, zwitterions, and drawing peptide (amide) linkages	☐	☐	☐
Protein structure and the effect of pH and temperature on enzymes	☐	☐	☐
Carbohydrates, triglycerides, saturated vs unsaturated fats and their properties	☐	☐	☐
Explaining solubility and melting point using intermolecular forces	☐	☐	☐

Polymers

Addition vs condensation polymerisation and identifying the repeating unit	☐	☐	☐
Structure–property relationships, including cross-linking and biodegradability	☐	☐	☐

WEAK SPOTS TO COME BACK TO

Topic 4 — Managing Resources

TOPIC 4 OF 4

Redox and electrochemistry

CONTENT	REVISED	PRACTISED	CONFIDENT
Assigning oxidation numbers and identifying oxidising and reducing agents	☐	☐	☐
Writing and balancing half-equations in acidic and basic conditions	☐	☐	☐
The metal activity series and predicting whether a reaction will occur	☐	☐	☐
Galvanic cells: electrode reactions, cell notation and calculating EMF	☐	☐	☐
Electrolytic cells, electroplating, and refining vs production of metals	☐	☐	☐
Faraday's laws and calculations involving charge, time and moles of electrons	☐	☐	☐

Fuels and energy

Combustion, calculating energy released in kJ g^{-1} and kJ L^{-1}	☐	☐	☐
Writing thermochemical equations with correct ΔH	☐	☐	☐
Calorimetry calculations ($Q = mc\Delta T$) and identifying sources of systematic error	☐	☐	☐
Biodiesel and ethanol production, and comparing fuels on environmental grounds	☐	☐	☐

Sustainability

Green chemistry principles and reducing environmental impact	☐	☐	☐
Distinguishing by-products from waste products	☐	☐	☐
Recycling, resource depletion and life-cycle considerations	☐	☐	☐

WEAK SPOTS TO COME BACK TO

WHAT TO DO WITH THE BLANKS

The gaps you just found are the whole point.

Every line you could not tick in the ‘Confident’ column is a mark you are currently at risk of dropping. Our SACE Exam Crash Course is built to close exactly those gaps — we mapped four years of real SACE papers to find where marks are actually lost, then weighted the teaching toward those points instead of re-covering content you already know.

The SACE Exam Crash Course — Chemistry

- 28 September – 10 October · Methods, Chemistry, Physics, Biology
- 12 hours of teaching per subject · online or in person, same live sessions
- TopMark Chemistry textbook included free
- Finishes with a full mock exam, marked and returned with your personal mark-loss breakdown
- 32 five-star Google reviews · 100% of our 2025 cohort scored above a 90 ATAR

From $349 for one subject, or $20/hr across all four. A $99 deposit secures your spot. Early bird pricing ends 6 September.

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